Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of professional experience building scalable backend services and high-impact web applications across healthcare, fintech, e-commerce, and civic tech. Deep expertise in **Node.js**, **TypeScript**, and **NestJS** for designing RESTful APIs, event-driven systems, and distributed architectures, complemented by strong **React** and **Next.js** frontend capabilities. Proven track record of taking full ownership of features from design through production deployment, implementing containerized CI/CD pipelines with **Docker**, and mentoring engineering teams to deliver reliable, well-tested software.

CORE COMPETENCIES

- **Backend & APIs:** NestJS, Node.js, TypeScript, RESTful API design, event-driven architecture (SSE, EventSource), JWT authentication, RBAC, and GraphQL.
- **Databases & ORM:** PostgreSQL (pgvector), MongoDB, Prisma, and database schema design.
- **Cloud & Infrastructure:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Traefik, and Nx monorepo orchestration.
- **Frontend:** React, Next.js, Vue.js, Redux, Zustand, Tailwind CSS, Radix UI, SSR/SSG, and performance optimization.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Mastra, vector search (pgvector, Chroma), and multi-agent workflows.
- **Leadership:** Technical ownership, architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, designing clean API interfaces for real-time conversational data querying.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling, collaborating closely with backend engineers to align on auth strategies and API contracts.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing adoption.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Optimized application performance by implementing lazy loading, code splitting, and CDN asset delivery, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Architected a reusable **design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume bank **Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**, processing payments across **MasterCard, VISA, Express-cash, and E-commerce** modules.
- Developed critical payment modules using **React**, **Redux**, **RxJS** for reactive data streaming, and **Axios** for asynchronous API communication with backend settlement services.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for **event-driven, real-time data streaming**, replacing conventional socket.io methods for faster data updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**, streamlining management of user roles and permissions.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** to power new features, translating business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, **NestJS** document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with **pnpm** workspaces.
- Designed and built backend services with **NestJS** and **TypeScript**, implementing clean API interfaces, **SSE streaming** for real-time chat responses with 15s keepalive heartbeats to prevent reverse proxy timeouts, and phone OTP authentication with session management.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling that filters out framework-managed tables from Prisma's drift detection.
- Built a **multi-agent AI system** using Mastra with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Traefik** for reverse proxy routing.
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management across environments, and built the Next.js citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts** featuring 22 interactive chart types.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*